

SYS-I0-AUX

Rev E0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | System I/O and auxiliary board, 25-110 kW DC/DC family (ECO-03/04/06/07)

01 24 V entry, protection and window

2 cabinet feeds (LM74700 ORing) + AUX-HV feed (LM74800 switch, cabinet priority, OV), 2 mF surge reservoir, precision OV cut-off 26.13-26.42 V, PG_24V window, eFuse UVLO

File: sheets/01_power_in.kicad_sch

02 24 V branches and local rails

+24V to CTRL-C2000 (2 A eFuse), +24V_GD gate-drive bias (5 A eFuse, channel B), 5 V buck (LMR36015), 3.3 V LDO, rail LEDs and test points

File: sheets/02_rails.kicad_sch

03 CTRL-C2000 and BMU-GW connectors

J301 = interfaces.SYS 2x40 (every contract signal), J302 = interfaces.GW 2x5; CANA and SCIC go to the BMU-GW socket; SCIC receiver always enabled

File: sheets/03_ctrl_if.kicad_sch

04 Isolated CAN ports (CANB, MCAN)

CANB module bus and MCAN CAN-FD service port; ISO1042 + SN6505B isolated 5 V each, jumper-selectable split termination, ESD2CAN24 TVS, 4-pole field terminal

File: sheets/04_can.kicad_sch

05 Isolated RS-485 ports (SCIA, SCIB)

Two half-duplex RS-485 ports; ISO1410 + SN6505B isolated 5 V each, jumper-selectable 120R termination, SM712 TVS, 4-pole field terminal

File: sheets/05_rs485.kicad_sch

06 Ethernet magnetics and chassis bond

RJ45 with integrated 1:1 magnetics for the 100BASE-TX PHY on CTRL-C2000, centre taps on ETH_CT (PHY analog supply), link / activity LEDs, PE bonding point

File: sheets/06_eth.kicad_sch

07 Isolated 24 V digital inputs

8 x IEC 61131-2 type 1/3 inputs, 4 x ISO1212; DI1 ESTOP_A (U701), DI8 ESTOP_B (U704) feed the chain; FB_A/B_MAIN (PELV GND) on J702 and in their own package U703

File: sheets/07_di.kicad_sch

08 Dual-channel safety chain and watchdog

A: E-stop A, 24 V UV, spare loop, watchdog -> latch A -> GATE_EN, 24V_DO switch A
B: E-stop B, watchdog -> latch B -> CHB_OK -> +24V_GD, 24V_DO switch B

File: sheets/08_chain.kicad_sch

09 Gated 24 V outputs (contactors, relays)

DO1..DO8 per interfaces.DO on 4 x TPS272C45D behind two series supply switches; switch A (channel A, DO_PWR_EN) and switch B (channel B, CHB_OK); TVS coil clamps

File: sheets/09_do.kicad_sch

10 Fan channels

4 x TPS16630 fan supply (3.6 A limit, auto-retry, OV cut-off), open-drain PWM, clamped tach; sized for PFC1224DE-F00 (2.0 A typ, 2.4 A max, 26.5 V max)

File: sheets/10_fans.kicad_sch

11 NTC inputs, ADC and supply monitors

ADS7953 16-ch SAR on SPL_CS_ADC_N, REF3025 2.5 V; CH0-11 = NTC1-12, CH12-14 = feed voltages, CH15 = +24V_GD gate-drive bias current (eFuse IMON)

File: sheets/11_ntc.kicad_sch

12 Service storage and diagnostics

I2C: FM24CL64B F-RAM 0x50 (serial, calibration, fault log, hours), TMP1075 0x48; SPL_CS_IO_N: two MCP23S17 (HAEN, A0 0/1): faults, rail sense, channel self-test

File: sheets/12_service.kicad_sch

24 V: two cabinet feeds (fused 30 A, TVS 33 V, LM74700 ideal-diode ORing) and the AUX-HV bootstrap feed (fused, TVS, LM74800 load switch with OV cut-off and a CdVdT ramp; it connects only when no cabinet feed is healthy) onto 24V_OR with a 2 mF surge reservoir. Every outgoing branch is a TPS16630 eFuse with the shared precision OV cut-off (26.13-26.42 V, D-013) and UVLO (16 V): +24V to CTRL (2 A), +24V_GD gate-drive bias (5 A, channel B), 24V_DO via switch A and switch B in series (6 A), four fans (3.6 A). While AUX-HV is the source, hardware forces the fans, +24V_GD and 24V_DO off and latches both chain channels (D-022: AUX-HV <= 30 W; checked every build). PG_24V = inside the 21.6-26.0 V window.

Safety chain (ECO-04), two channels on separate parts: A = E-stop A, 24 V UV, spare loop, watchdog -> latch A -> GATE_EN + 24V_DO switch A; B = E-stop B, watchdog -> latch B -> CHB_OK -> +24V_GD + 24V_DO switch B. Each channel alone stops the gates and opens 24V_DO; the build simulates every single stuck fault (outputs/SYS-I0-AUX_safety_fmea.csv).

Isolation: CANB, MCAN, RS-485 A/B each have their own SN6505B supply and ground; the DI field side (DIF_GND) is a separate domain; PE is reached only through the RJ45 Bob-Smith node and the 470 pF DI field-ground capacitor. Budget: 16 A steady worst case, 20 A for 100 ms.

Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.
Sheet 1
Schematics authored by Chinmoy Bhuyan
Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE - only isolators cross (checked)

Sheet: /
File: SYS-I0-AUX.kicad_sch

Title: SYS-I0-AUX - Cover / Hierarchy

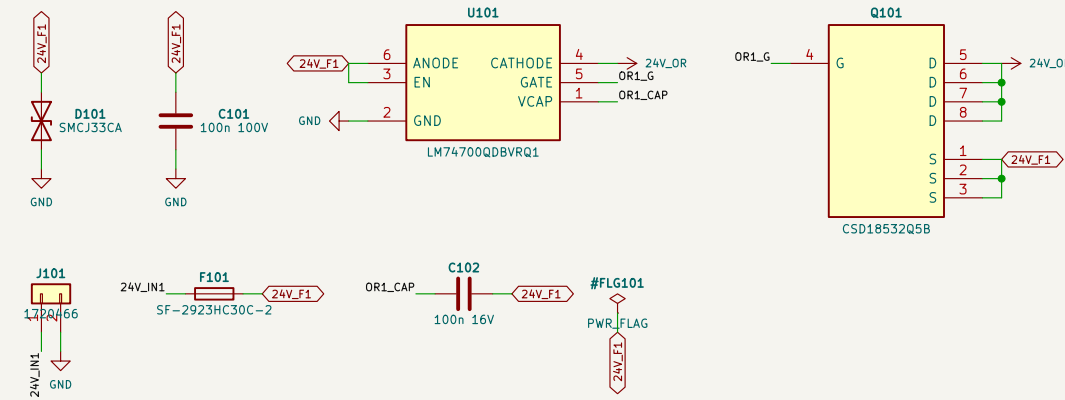
Size: A3 | Date: 2026-10-04

Rev: E0

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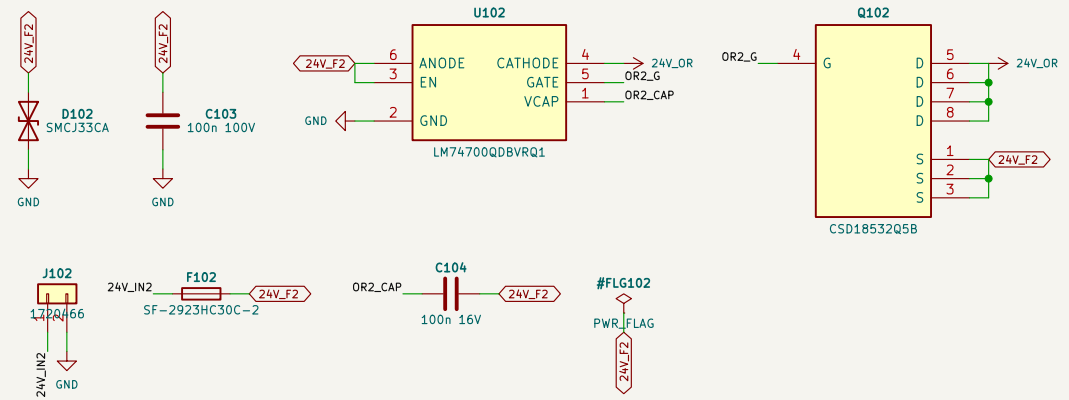
Id: 1/13

FEED 1: CABINET 24 V A



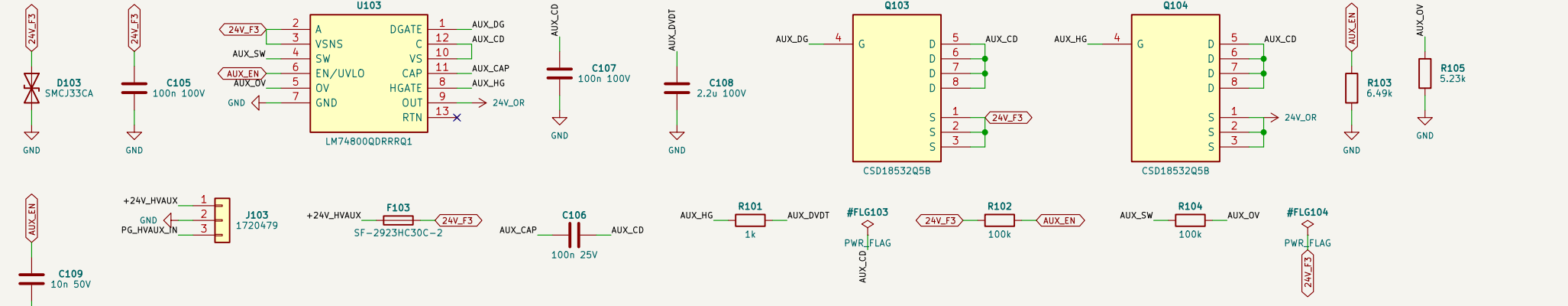
Fuse 30 A / 60 VDC: TVS 1500 W bidirectional (53 V clamp, blocks -33 V wiring error).
LM74700 + CSD18532Q5B: ideal diode, ORs the feeds, blocks reverse current and polarity.

FEED 2: CABINET 24 V B



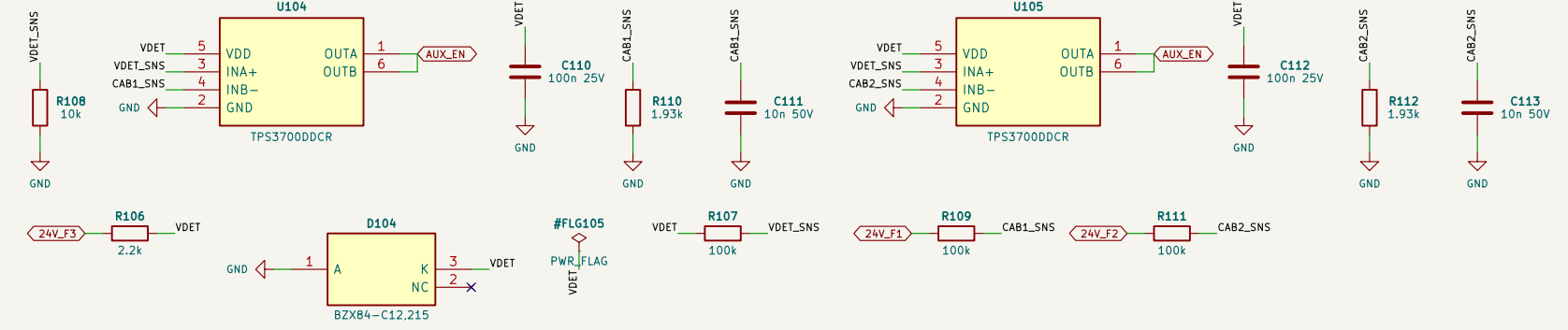
Fuse 30 A / 60 VDC; TVS 1500 W bidirectional (53 V clamp, blocks -33 V wiring error).
LM74700 + CSD18532Q5B: Ideal diode, ORs the feeds, blocks reverse current and polarity.

FEED 3: AUX-HV 24 V (30 W BOOTSTRAP, INTERFACES.AUX)



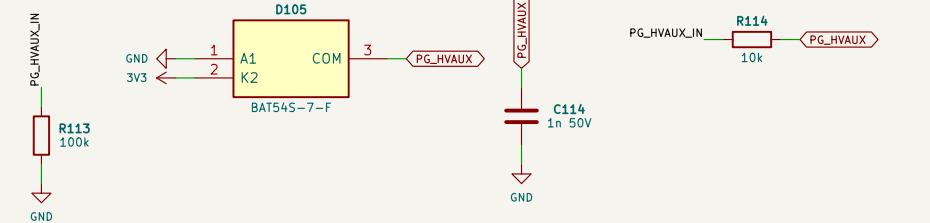
LM74800 + 2 x CSD18532Q5B back-to-back (common drain); ideal diode (DGATE) + load disconnect (HGATE).
 Connects only when NO cabinet feed is healthy (priority detector holds EN/UVLO in shutdown); cabinet healthy
 above 20.85 - 21.41 V rising, lost below 20.37 - 21.20 V falling (only in window: cabinet wins, INT - I/O). AUX UVLO
 20.9 - 21.49 V rising / 18.54 - 19.46 V falling; AUX OVS 25.97 - 25.57 V, re-arm below 21.98 - 23.39 V; failed AUX-HV
 (23.8 - 28.5 V) is cut off before the 26.13 V bus cut-off and stays off (INT-OV). AUX-HV overshoot must stay <
 23.97 V, while AUX-EN is in, sheet 0B forces fans + 24V_DO and 24V_DO off, latched (D-022, < 30 W).
 HGATE ramp (LM74800 + 10 nF to 50 nF, R5 at 1 kOhm 220 Ohm VDR; a disabled HGATE sits at OUT, so COLD tracks
 the bus. Black start: 0.016 - 0.063 V/ms, one ramp in 0.39 - 1.53 s, AUX-HV < 36.0 V (limit 39.8 V), < 5 W for < 60
 ms (liner > 88 ms). Cabinet -> AUX: Inhibit < 0.17 ms, HGATE on < 1.08 ms, FEI conducts < 5.98 ms after
 24V_DO is asserted; bus -> INT 2 V > logic UVLO 15.48 V (+24V fuse) at 27.8 V; AUX -> cabinet: AUX-HV off
 -35 us, bus fails to the cabinet - no gap; +24V_DO / 24V_DO stay until FLICLR.

CABINET PRIORITY DETECTOR (POWERED FROM THE AUX FEED)



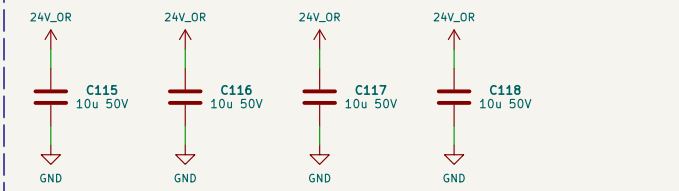
BZX84-C12 rail from 24V_F3; works before any logic rail exists, so AUX-HV can start the module alone. Each TPS3700 OUTB pulls AUX_EN low (LM74800 shutdown, both FETs off) while its cabinet feed is healthy. OUTA (INA+ = VDET / 11) holds AUX_EN low while the detector itself is starting (SBVS187G 7.4.2, fig 22), so a hot-plugged AUX-HV can neither enable the LM74800 nor trip the inhibit before the decision is valid.

AUX-HV POWER-GOOD



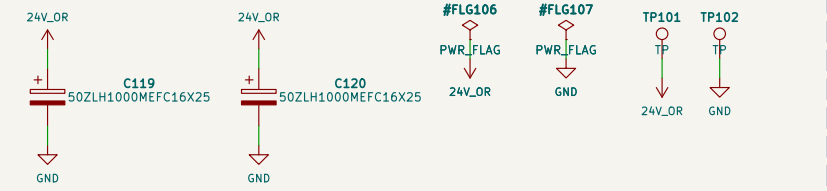
PG_HVAUX is driven actively high by AUX-HV when good (interfaces.AUX): 100k pull-down here, so a dead or unplugged AUX-HV reads not-good. 10k + BAT54S clamp toward CTRL.

ORED 24 V BUS 24V_OR



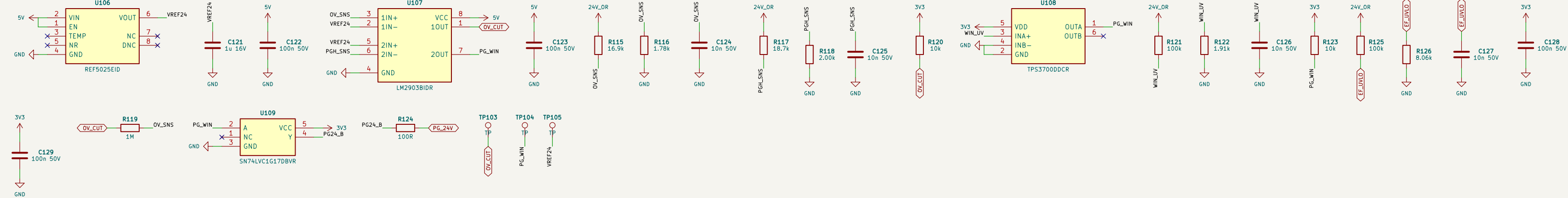
Cabinet feeds share by ideal-diode ORing; AUX-HV joins only through its LM74800 switch.

SURGE RESERVOIR



2 x 1000 uF 50 V low-impedance electrolytic absorb the surge charge (check surge, every build).
Design surge IEC 61000-4-5 0.5 kV / 12 ohm / 141.7 A, Q 3.00 m into 2 x 1000 uF (C_min 1.28 mF, ESR_max 32 mOhm)
-> bus and every branch can = 30.07 V [TVS idle below 36.7 V; worst load margin 1.9 V, UCC14243 32 V].
0.5 kV / 2 ohm line-to-line (EN IEC 61000-6-2 only for 24 V cables > 30 m) would reach 39 V: NOT covered -
keep the cabinet 24 V wiring < 30 m or fit an external SPD. The OV cut-off may open the eFuses for -1.3 ms.

24 V WINDOW: OV CUT-OFF, PG_24V, EFUSE UVLO



D-013: window 21.6–26.0 V; nothing downstream above 26.45 V steady. Worst-case stacks (corners of every trace)
OV cut-off = LM2930B-1 vs REF502SE: ref 2.5 V +/-0.066 % (0.025 % + 2.5 ppm/K x 165 K). VIO +4/-mV, IB 50 nA,
16.9k Ω /1.8k Ω O1, 10 ppm/kHz x 65 K each. 1 m hysteresis to VOL: 0-0.55 V -> 26.131-26.417 V -> OV,CUT -> OV all
8 efuses (85–14 us), PC2+M, LM2930B-2 18.7k/200k 20.740-26.012 V AND low effuse TPS3D07 INA:
1.09-1.105 mV, 100k/1.91k 20.538-21.4 k V. Chn UV (sheet 08) 1.09-1.105 mV, 100k/1.91k 20.538-21.4 k V. Chn UV (TPS1602 1.176-1.224 v,
1.09-1.15 mV, 100k/8.00k O1, 15.72-16.48 V on, 14.57-15.48 V off; +24V (CTRL) is the logic input at a hand-over,
TSN SMCJ33CA (VBR = 36.7 V) stays idle in the design surge - see the surge reservoir.

Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.
Sheet 2
Schematics authored by Chinmoy Bhuyan
Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE - only isolators cross (checked)

Sheet: /01 24 V entry, protection and window/
File: 01_power_in.kicad_sch

Title: SYS-10-AUX - 24 V entry, protection and window

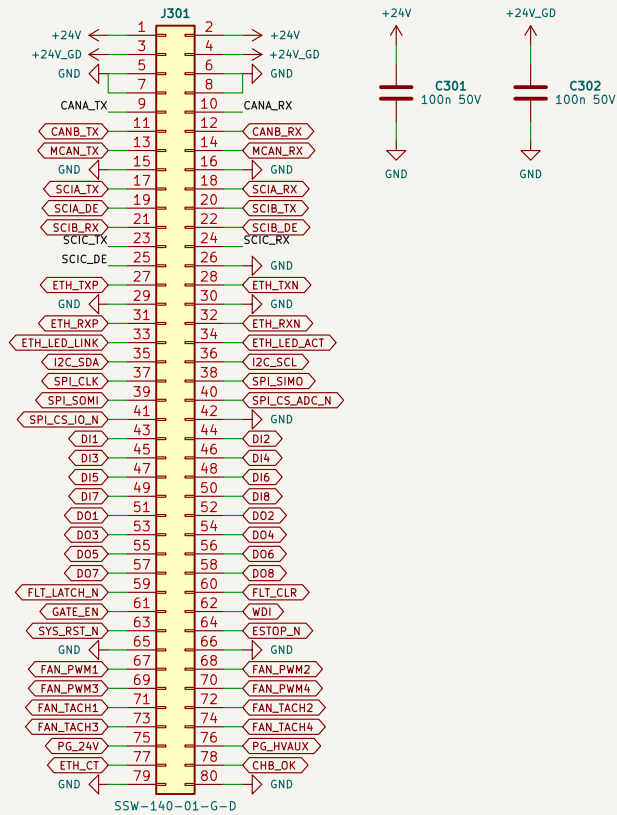
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Rev: E0

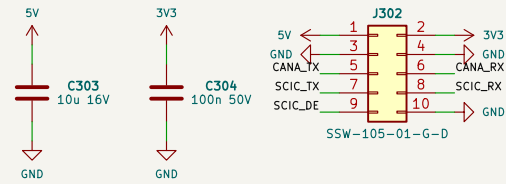
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CTRL-C2000 BOARD-TO-BOARD (INTERFACES.SYS)



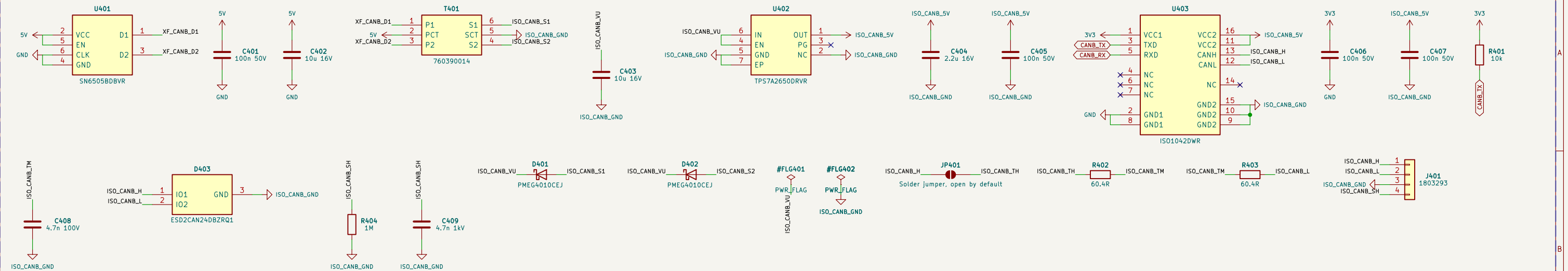
Samtec SSW-140-01-G-D socket mates the TSW-140-07-G-D header on CTRL (both catalogs list the pair; TSW -07 post 5.84 mm inside SSW insertion depth 3.68-6.35 mm). 4.7 A/pin; +24V and +24V_GD 2 pins each. No I2C pull-ups here (CTRL has them). Pull-ups / pull-downs sit next to each driver on its sheet.

BMU-GW SOCKET (INTERFACES.GW)



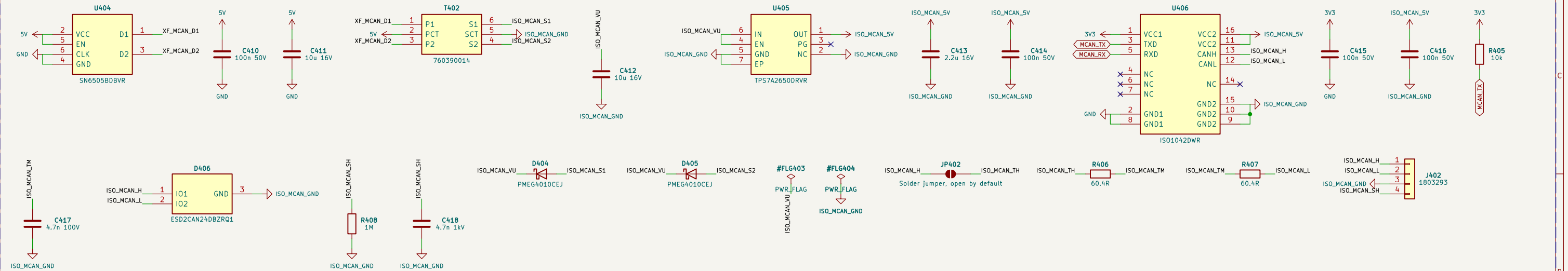
CANA -> CAN_TX/RX, SCIC -> RS485_D/R/DE; RS485_RE_N tied low here: the SYS contract carries no SCIC_RE, so the receiver stays on.

CANB MODULE BUS (ISOLATED)



ISO1042 5 kVrms, own isolated 5 V. Split termination 2 x 60.4R + 4.7n, closed by JP.
Terminal: 1 CANH, 2 CANL, 3 isolated GND, 4 shield. Port supply 760390014 + TPS7A2650: 5.0 V +/-1 %.

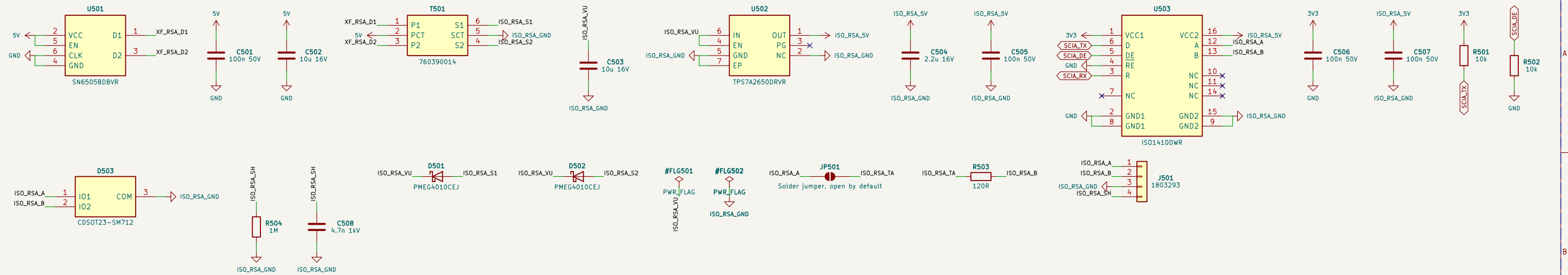
MCAN CAN FD SERVICE PORT (ISOLATED)



ISO1042 5 kVrms, own isolated 5 V. Split termination 2 x 60.4R + 4.7n, closed by JP.
Terminal: 1 CANH, 2 CANL, 3 isolated GND, 4 shield. Port supply 760390014 + TPS7A2650: 5.0 V +/- 1 %.

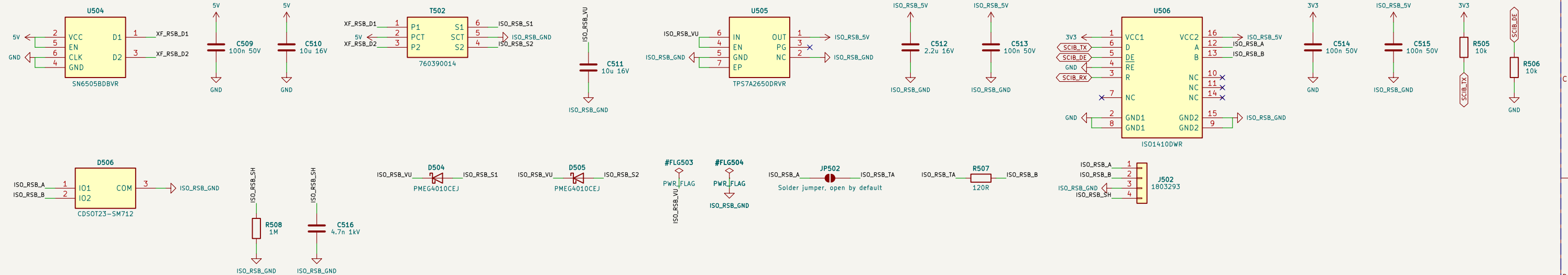
Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build. Sheet 5		
Schematics authored by Chinmoy Bhuyan Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE – only isolators cross (checked)		
Sheet: /04 Isolated CAN ports (CANB, MCAN)/ File: 04_can.kicad_sch		
Title: SYS-10-AUX – Isolated CAN ports (CANB, MCAN)		
Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 5/13

RS-485 PORT 1 = SCIA (ISOLATED)

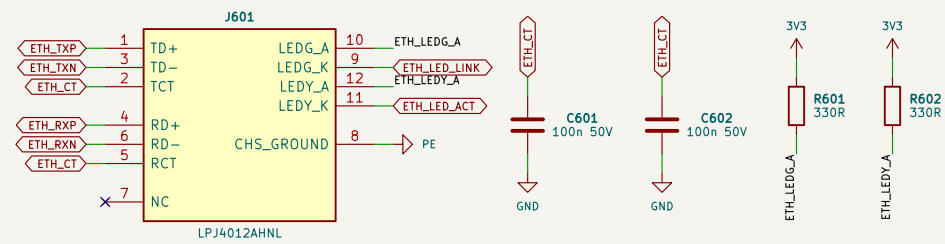


ISO1410 5 kVrms half duplex, own regulated isolated 5.0 V, fail-safe receiver always on (-RE = GND).
120R termination by JP. Terminal: 1 A, 2 B, 3 isolated GND, 4 shield (1M || 4.7n).

RS-485 PORT 2 = SCIB (ISOLATED)

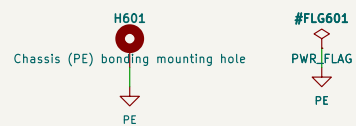


ISO1410 5 kVrms half duplex, own regulated isolated 5.0 V, fail-safe receiver always on (-RE = GND).
120R termination by JP. Terminal: 1 A, 2 B, 3 isolated GND, 4 shield (1M || 4.7n).



LINK-PP LPJ4012AHNL: 1CT:1CT, 1500 Vrms hipot, -40...+85 C (rev E, PV-20; was WE 7499010211A 0..70 C). Centre taps TCT / RCT on ETH_CT = the PHY's AVD sent from CTRL (SYS pin 77); decoupled only, no local 3.3 V. LEDs (VF 1.8-2.6 V); PHY pins are active-low (CTRL 2.49k strap pull-ups); 330R from local 3V3.

CHASSIS (PE) BOND



Plated mounting hole to the module chassis: RJ45 Bob-Smith node, DI field-ground 470 pF (sheet 07). PELV 0 V is not bonded here.

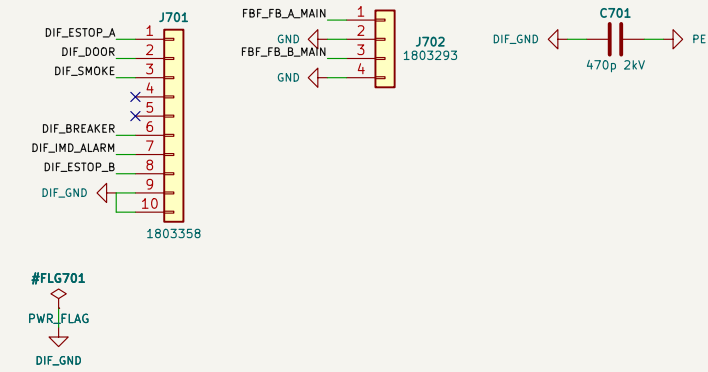
Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.
Sheet 7
Schematics authored by Chinmoy Bhuyan
Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE – only isolators cross (checked)

Sheet: /06 Ethernet magnetics and chassis bond/
File: 06_eth.kicad_sch

Title: SYS-IO-AUX - Ethernet magnetics and chassis bond

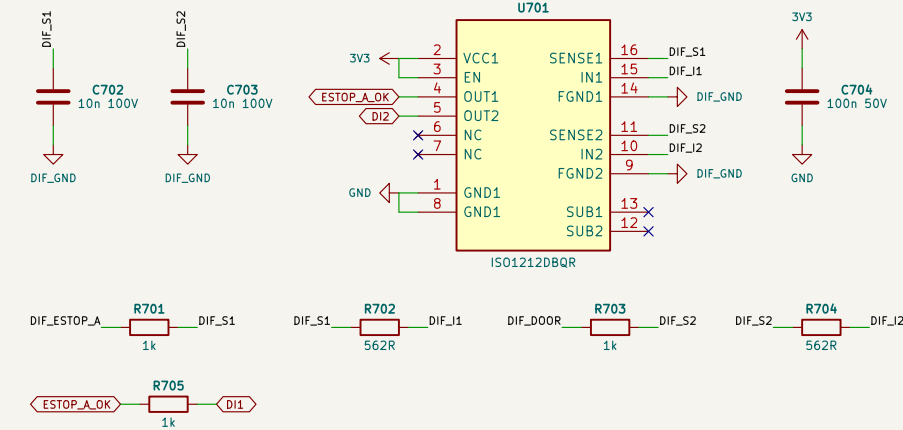
Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 7/13

DI FIELD TERMINALS (INTERFACES.DI)



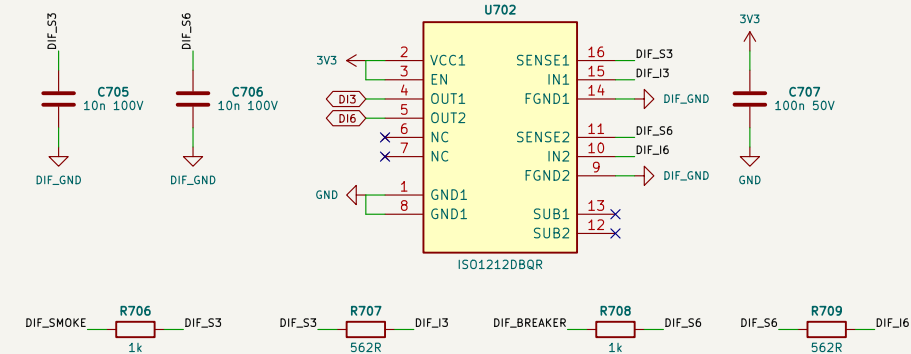
J701 pin n = Din for the six external inputs (4, 5 unused), 9/10 = field 0 V DIF_GND (470 pF / 2 kV to PE).
 FB_A_MAIN (DI4) / FB_B_MAIN (DI5) are module-internal, PELV: J702 1 FB_A, 2 GND, 3 FB_B, 4 GND.
 ISO1212 channels of one package share only functional insulation (+/-60 V, SLLSEY7G 6.1), so DI4 + DI5
 sit together in U703 with both FGND on GND; U701 / U702 / U704 carry only DIF_GND channels.

DI1 ESTOP_A, DI2 DOOR



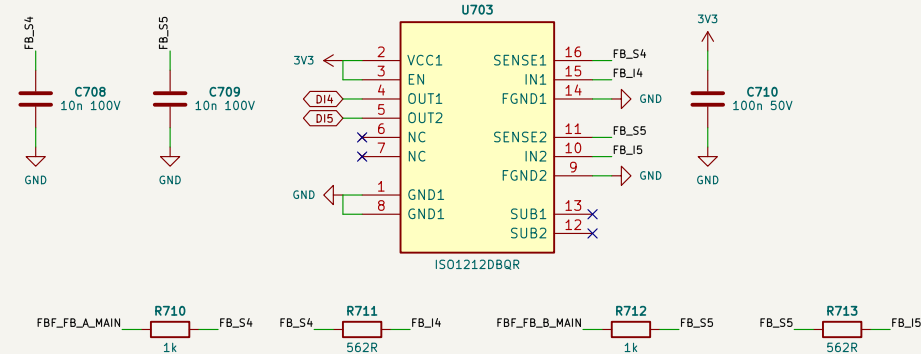
RTHR 1k MELF 0.25 W, RSENSE 562R, CIN 10n: VIH <= 10.95 V, VIL >= 8.7 V, 2.25 mA -> type 1 and 3.
SUB pins go to a floating 2 x 2 mm copper island only (datasheet), never to FGND. FGND = DIF_GND.
DI1 = E-stop channel A: drives the chain directly; DI1 to CTRL through 1k.

DI3 SMOKE, DI6 BREAKER



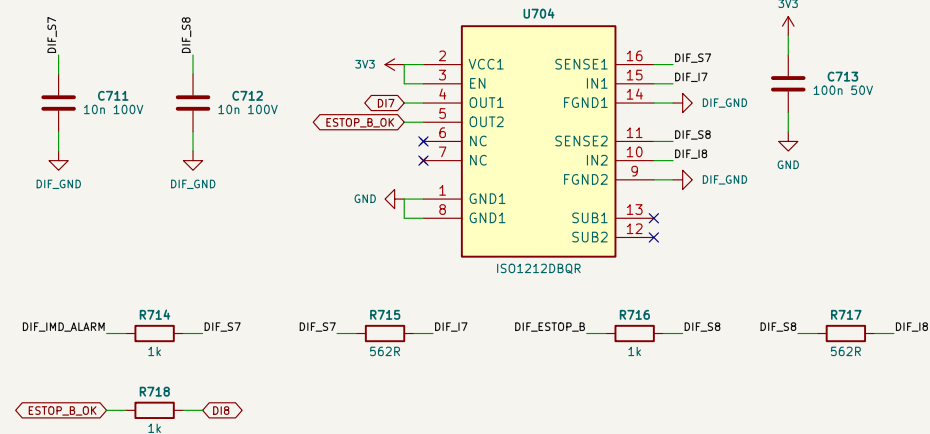
RTHR 1k MELF 0.25 W, RSENSE 562R, CIN 10n: VIH <= 10.95 V, VIL >= 8.7 V, 2.25 mA -> type 1 and 3.
SUB pins go to a floating 2 x 2 mm copper island only (datasheet), never to FGND. FGND = DIF_GND.

DI4 FB_A_MAIN, DI5 FB_B_MAIN



RTHR 1k MELF 0.25 W, RSENSE 562R, CIN 10n: VIH ≤ 10.95 V, VIL ≥ 8.7 V, 2.25 mA $\rightarrow$ type 1 and 3.
SUB pins go to a floating 2 x 2 mm copper island only (datasheet), never to FGND. FGND = GND.

DI7 IMD_ALARM, DI8 ESTOP_B



RTHR 1k MELF 0.25 W, RSENSE 562R, CIN 10n: VIH <= 10.95 V, VIL >= 8.7 V, 2.25 mA -> type 1 and 3.
SUB pins go to a floating 2 x 2 mm copper island only (datasheet), never to FGND. FGND = DIF_GND.
D18 = E-stop channel B: drives the chain directly; D18 to CTRL through 1k.

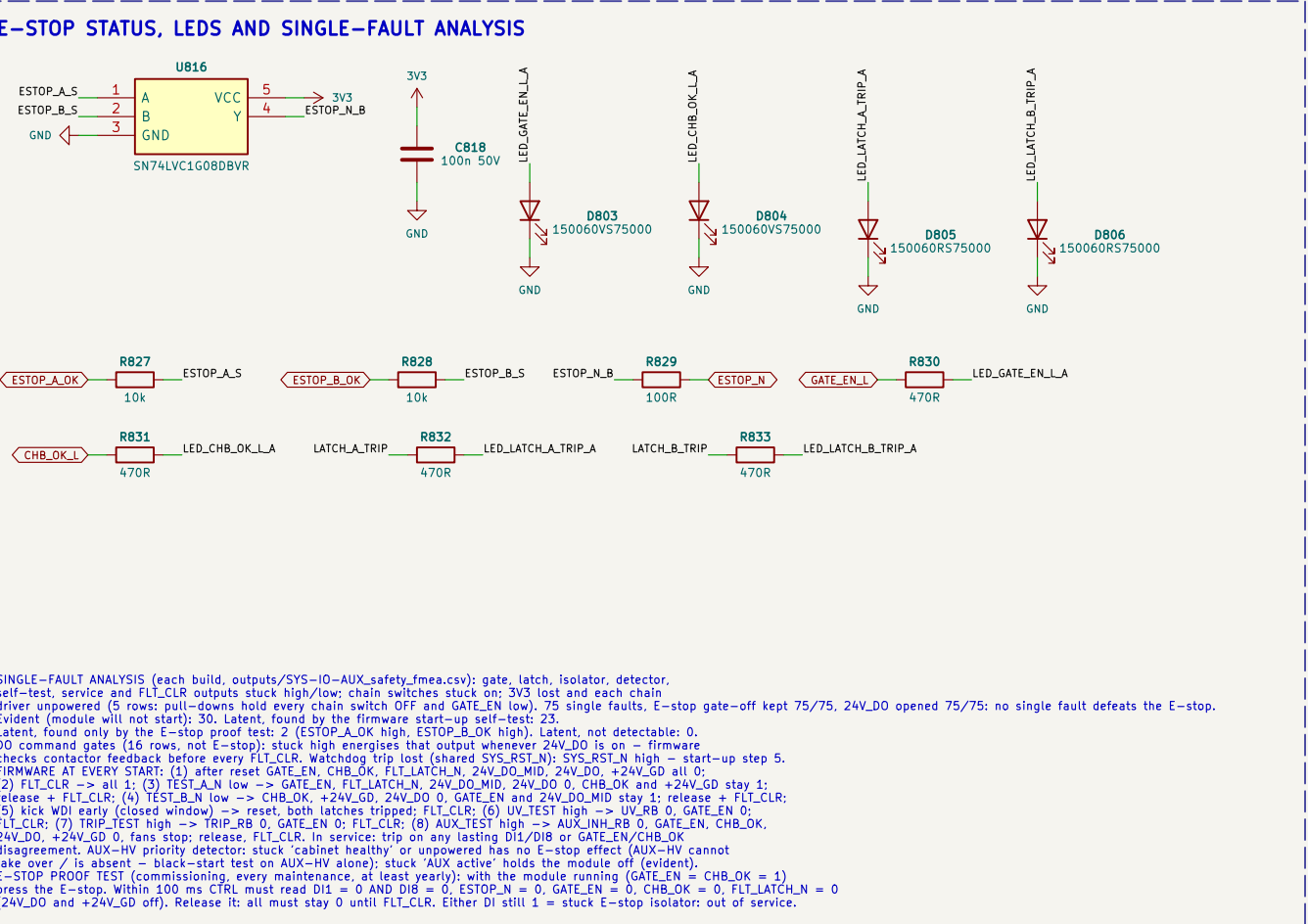
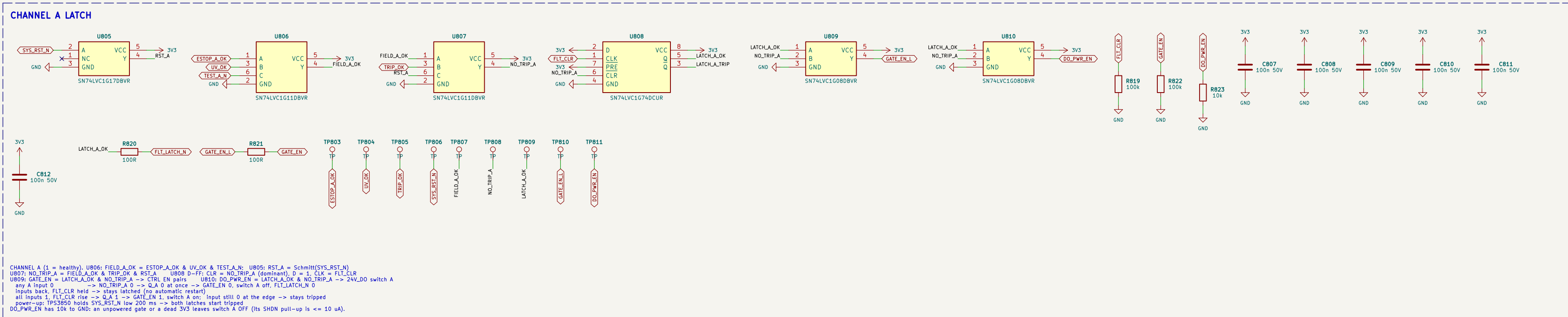
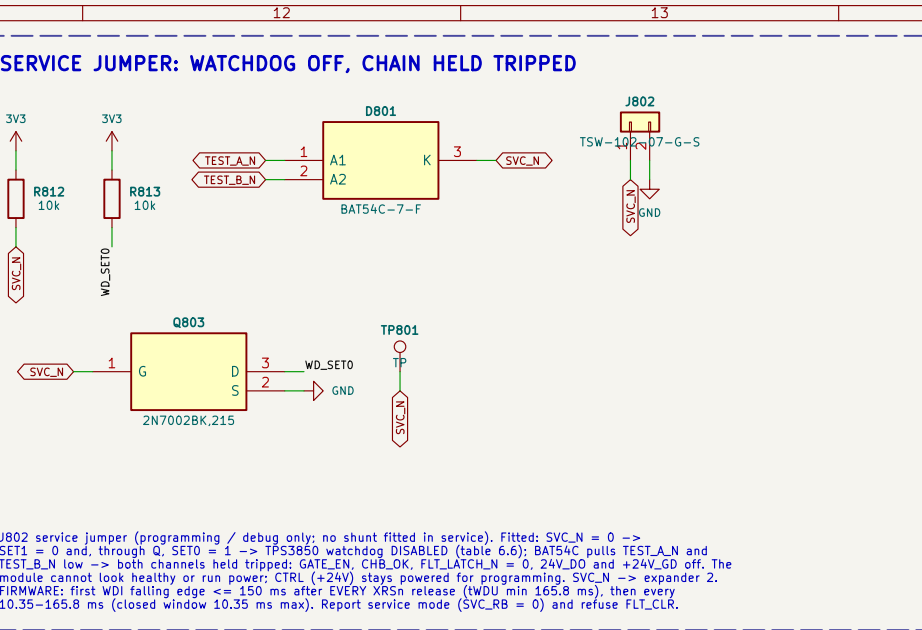
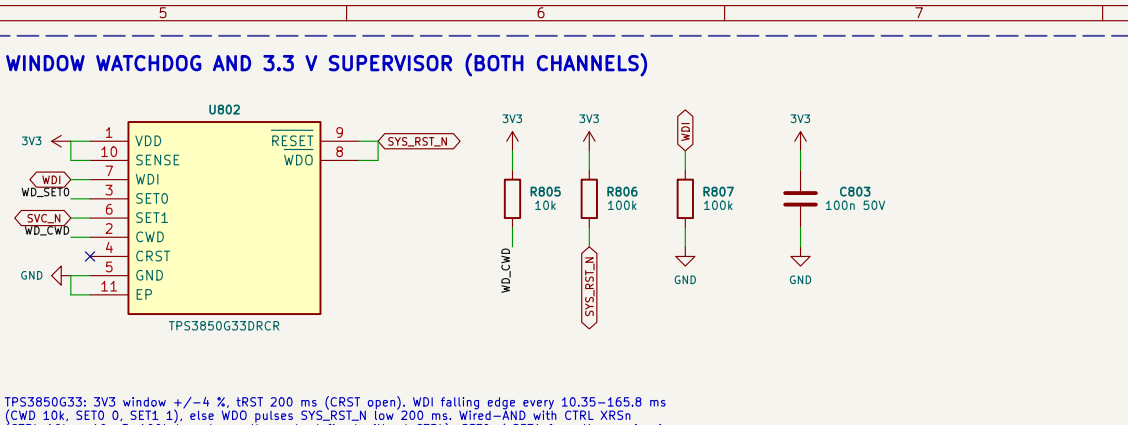
Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.
Sheet 8
Schematics authored by Chinmoy Bhuyan
Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE – only isolators cross (checked)

Sheet: /07 Isolated 24 V digital inputs/
File: 07_di.kicad_sch

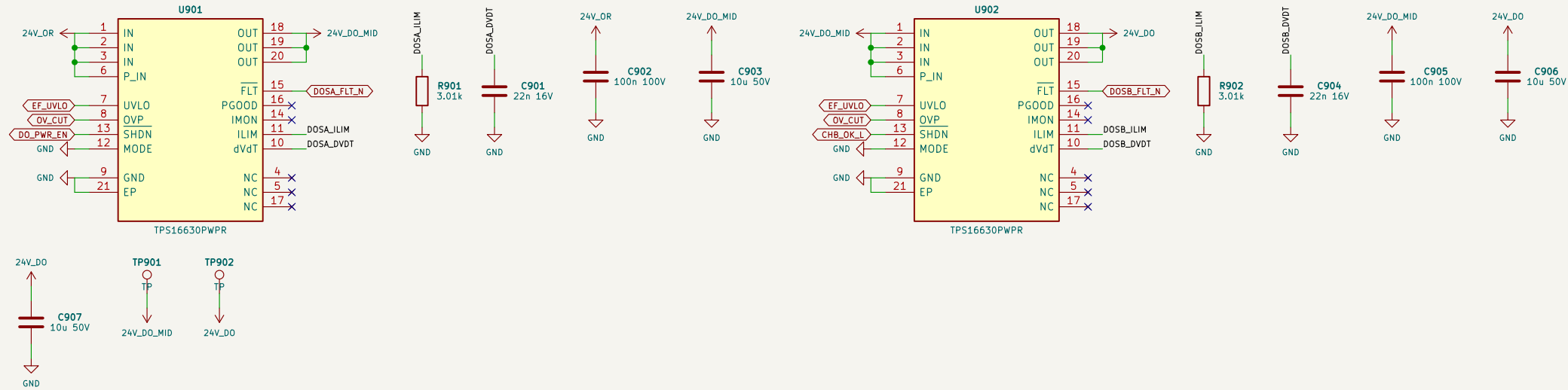
Title: SYS-10-AUX – Isolated 24 V digital inputs

Size: A3	Date: 2026-10-04
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Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 8/13

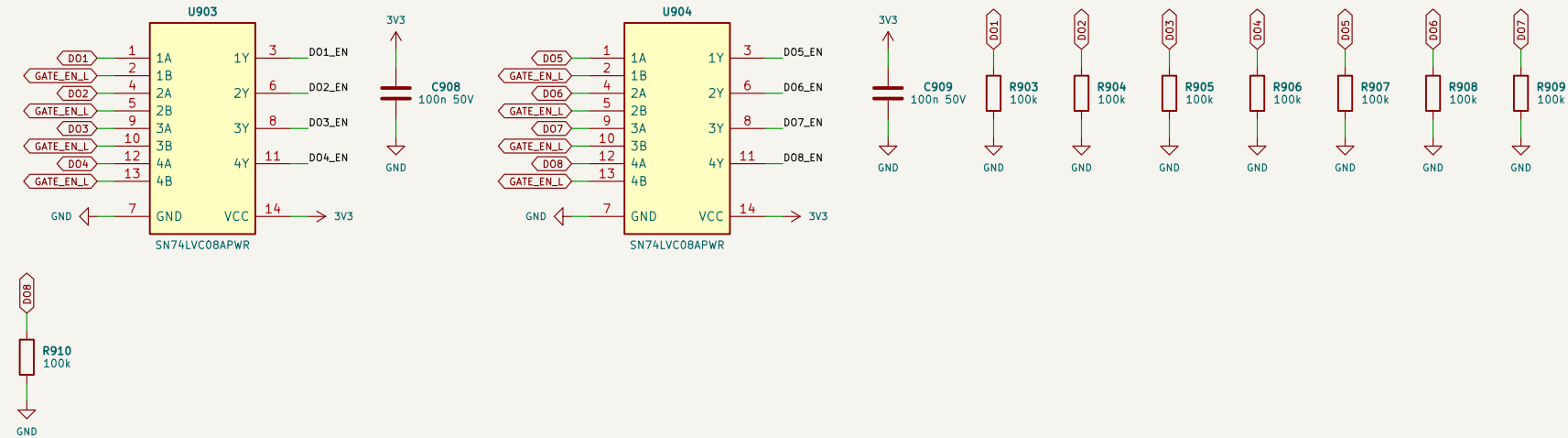


OUTPUT SUPPLY 24V_DO: TWO SERIES SWITCHES, ONE PER SAFETY CHANNEL



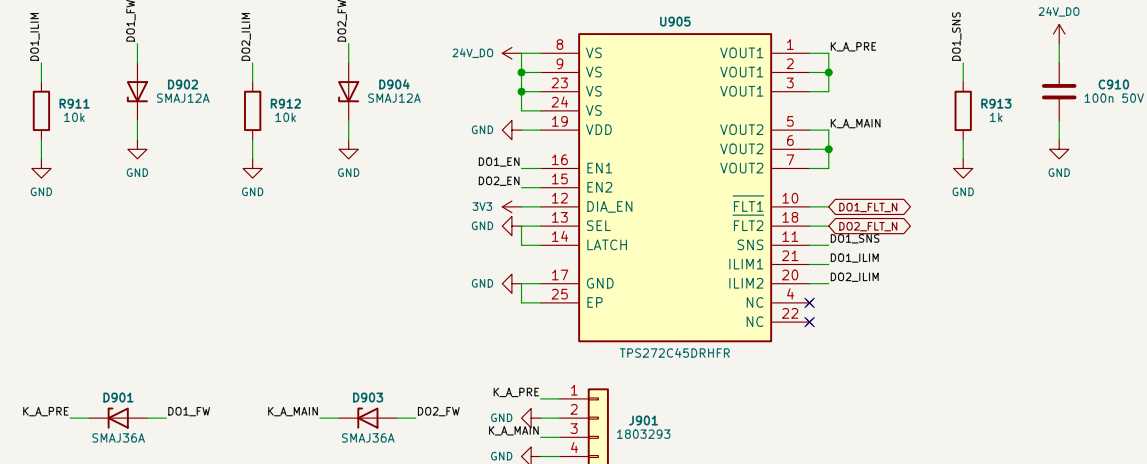
TPS16630 A (SHDN = DO_PWR_EN) -> 24V_DO_MID -> TPS16630 B (SHDN = CHB_OK_L) -> 24V_DO_ILM 3.01k -> 5.98 A (5.58-6.42 A) each. All eight outputs on: <= 1.51 A (two HVCA42 coll pairs, not economised); one output shorted: <= 3.81 A - the other outputs keep running (check_do_loads). Rails read back via expander 2. While AUX-HV is the source both switches are held off (sheet 08).

COMMAND GATING



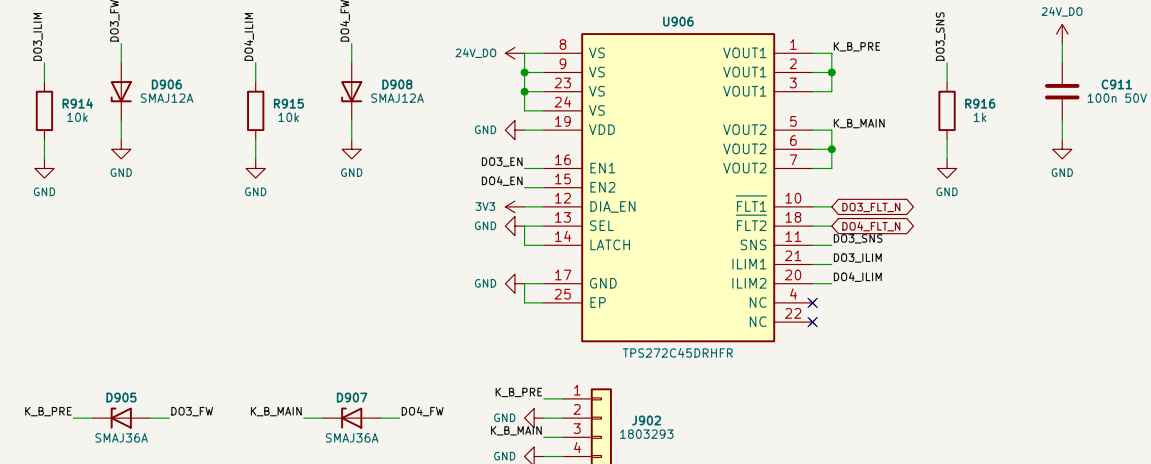
DOn_EN = DOn & GATE_EN (SN74LVC08A). 100k pull-downs: outputs off while CTRL is absent or in reset. K_DISCH (D05) is held on only while both channels are healthy.

D01 K_A_PRE, D02 K_A_MAIN



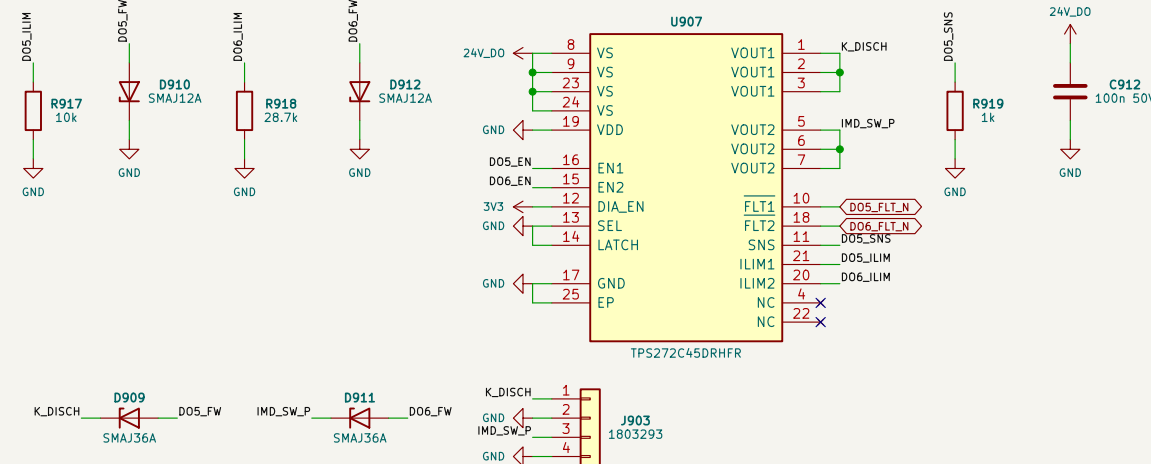
Port A precharge + main controller. FLT1/2 -> expander 1. Clamp SMAJ36A (blocks +36 V) + SMAJ12A: OUT >= -15.7 V. Port board (interfaces_D0): the K_{XX} MAIN line feeds the coil AND gates its low-side switch: the coil's >= 50 V varistor is its only flyback path. At turn-off the gate falls with this output, the coil current circulates in coil + varistor and this pin only pins the loop: VS - VOUT <= 42.1 V < 48 V abs max (check_coilTurnoff). Battery-type port (DAB both, PV-PORT B): TWO HVC43 coils in parallel, <= 0.60 A, not economised (pull-in = hold) -> I_{ILM} 10k = 1.58-2.30 A (check_coil_loads).

D03 K_B_PRE, D04 K_B_MAIN



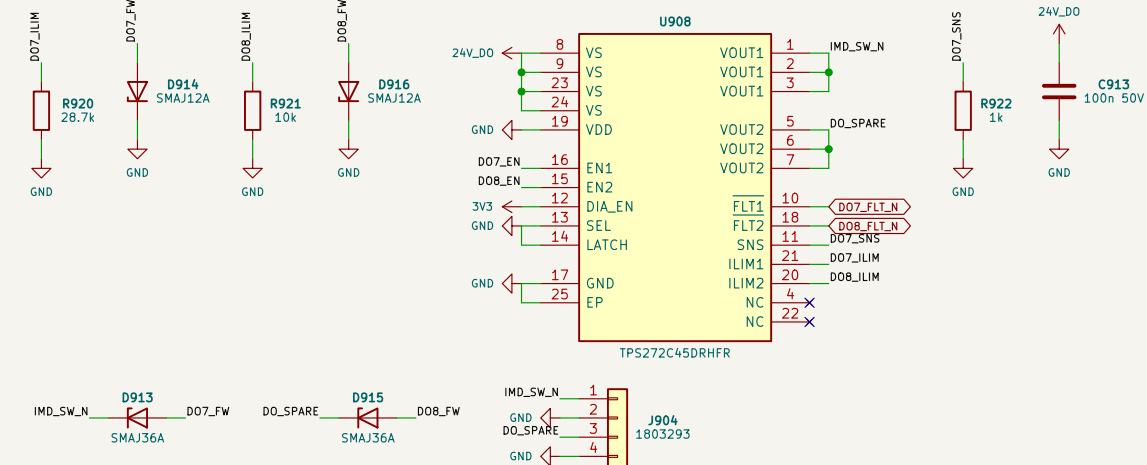
Port B precharge + main controller, FLT1/2 -> expander 1. Clamp SMAJ36A (blocks +36 V) + SMAJ12A: OUT >= -15.7 V. Port board (interfaces DO): the K_X_MAIN line feeds the coil AND gates its low-side switch; the coil's >= 50 V varistor is its only flyback path. At turn-off the gate falls with this output, the coil current circulates in coil + varistor and this pin only pins the loop: V5 - VOUT <= 42.1 V < 48 V abs max (check_coil_turnoff). Battery-type port (DAB both, PV-PORT B): TWO HVC43 coils in parallel, <= 0.60 A, not economised (pull-in = hold) -> ILIM 10k = 1.58-2.30 A (check_dloads).

D05 K_DISCH, D06 IMD_SW_P



Discharge relay + IMD switch P. FLT1/2 -> expander 1. Clamp SMAJ36A (blocks +36 V) + SMAJ12A; OUT ≥ -15.7 V.

D07 IMD_SW_N, D08 DO_SPARE



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MER switch N + spare, FL1/2 -> expander I, Clamp SMA356 (blocks +36 V) SMA12A: OUT ~ -15.7 V.
PER-OUTPUT CAPACITANCE (85 C ambient; calculated, not tested). Clamp energy every output: 0.5 J per turn-off, 0.5 W average (SMA12A 400 W/1 ms, -150 W/10 ms, derated to 85 C).
D01_KA_PARE ILM 10k 1.58-2.30 A cont 1.5 A Irush 1.5 A x unlimited (<= continuous)
D02_KA_MAIN ILM 10k 1.58-2.30 A cont 1.5 A Irush 1.5 A x unlimited (<= continuous)
D03_KB_PRE ILM 10k 1.58-2.30 A cont 1.5 A Irush 1.5 A x unlimited (<= continuous)
D04_KB_MAIN ILM 10k 1.58-2.30 A cont 1.5 A Irush 1.5 A x unlimited (<= continuous)
D05_KDISCH ILM 10k 1.58-2.30 A cont 1.5 A Irush 1.5 A x unlimited (<= continuous)
D06_IDW_SPARE ILM 287k 0.52-0.82 A cont 0.5 A Irush 0.5 A x unlimited (<= continuous)
D07_SW_SPARE ILM 287k 0.52-0.82 A cont 0.5 A Irush 0.5 A x unlimited (<= continuous)
D08_D0_SPARE ILM 10k 1.58-2.30 A cont 1.5 A Irush 1.5 A x unlimited (<= continuous)

```

Not bench-validated. Dual-channel chain gate- and single-fault-checked on every build.
Sheet 10
Schematics authored by Chinmoy Bhuyan
Domains: PELV, CANB, MCAN, RSA, RSB, DI field, PE - only isolators cross (checked)

Sheet: /09 Gated 24 V outputs (contactors, relays)/
File: 09_do.kicad_sch

Title: SYS-10-AUX - Gated 24 V outputs (contactors, relays)

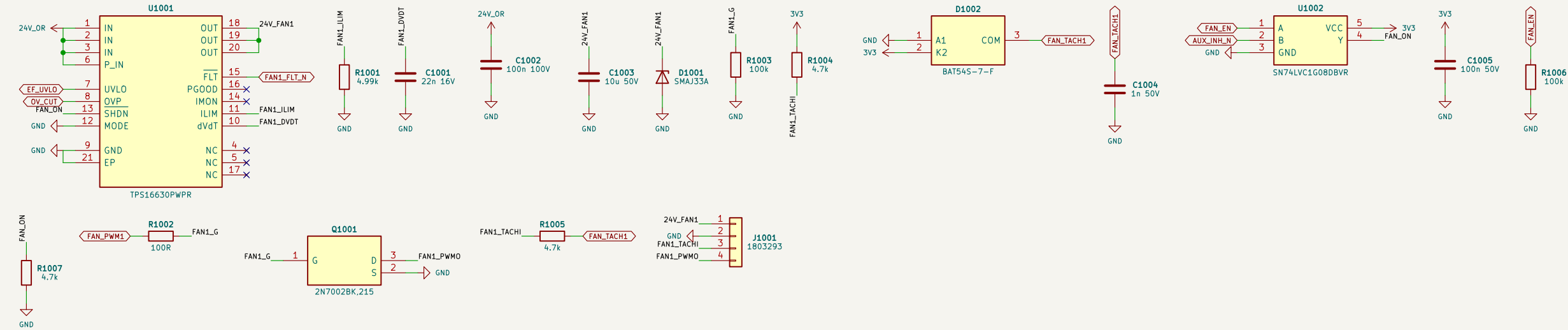
Size: A2	Date: 2026-10-04
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KiCad E.D.A. 10.0.6	Id: 10/13
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10

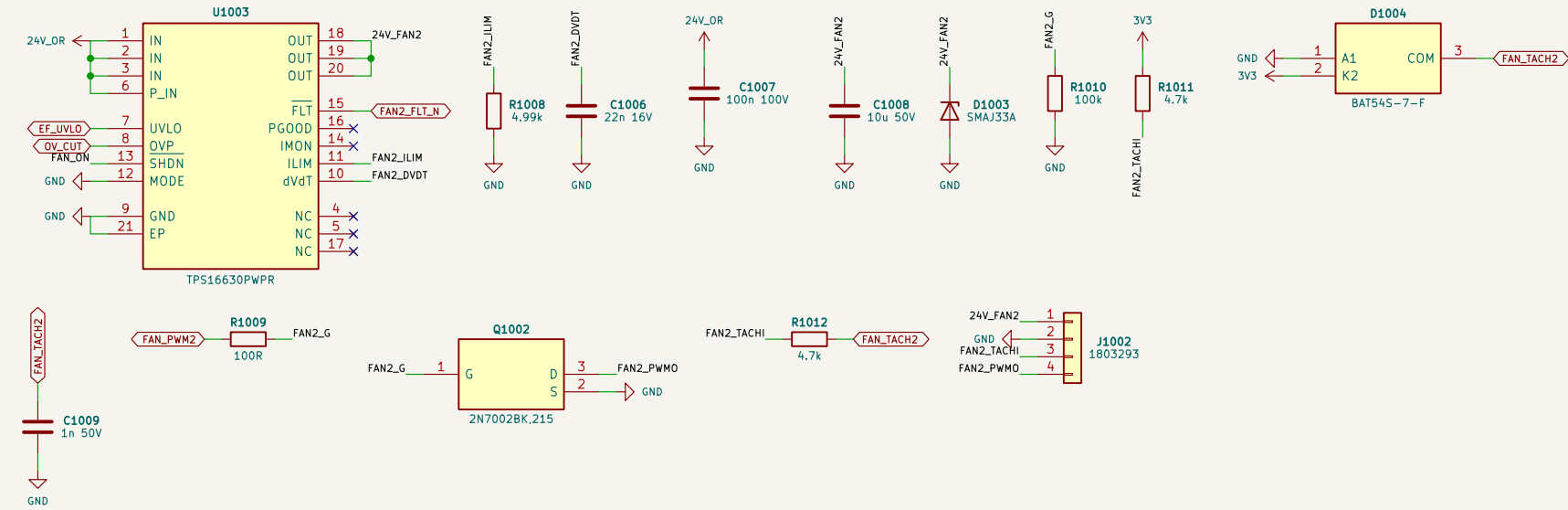
1	
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FAN 1

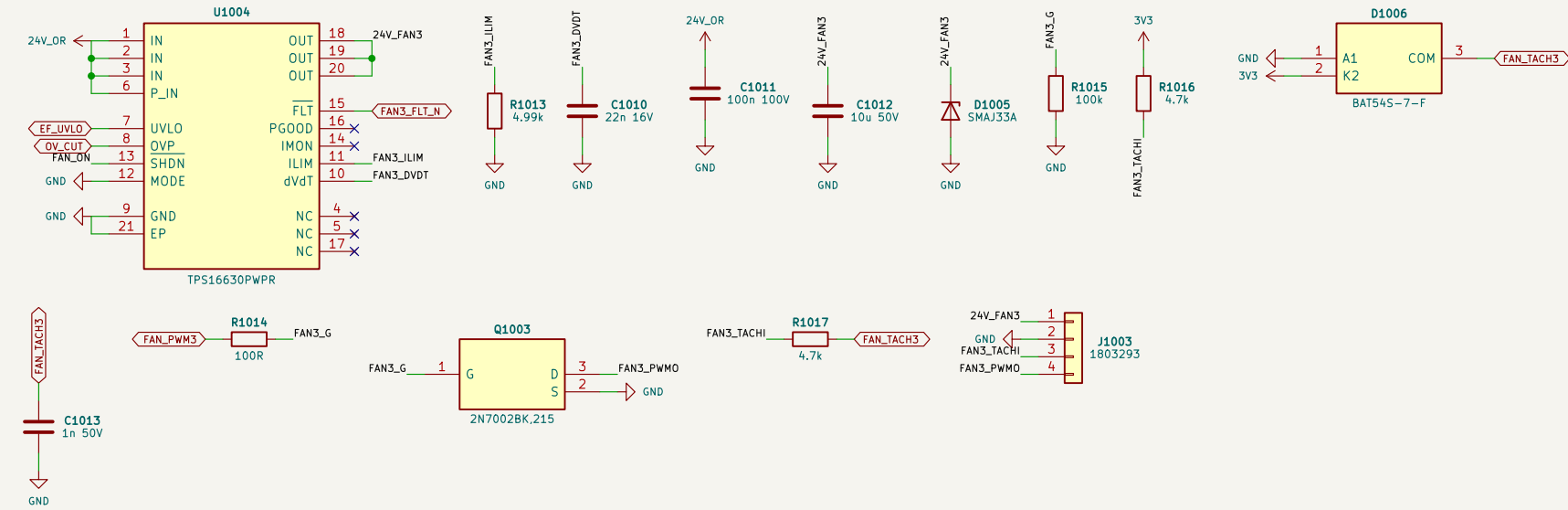


Terminal: 1 +24 V, 2 0 V, 3 tach (blue), 4 PWM (yellow). ILIM 4.99k -> 3.6 A.
FAN_ON = FAN_EN (expander 1 GPB4, 100k pull-down) AND AUX_INH_N: fans are OFF until
firmware sets FAN_EN and whenever AUX-HV is the source (no firmware in that path);
4.7k on FAN_ON keeps all four off if the gate is unpowered.
PWM inverted: FAN_PWM high = fan input low; once powered, PWM open = full speed.
Tach: open collector, 4.7k to 3V3 (0.7 mA < 5 mA), 4.7k + BAT54S clamp.

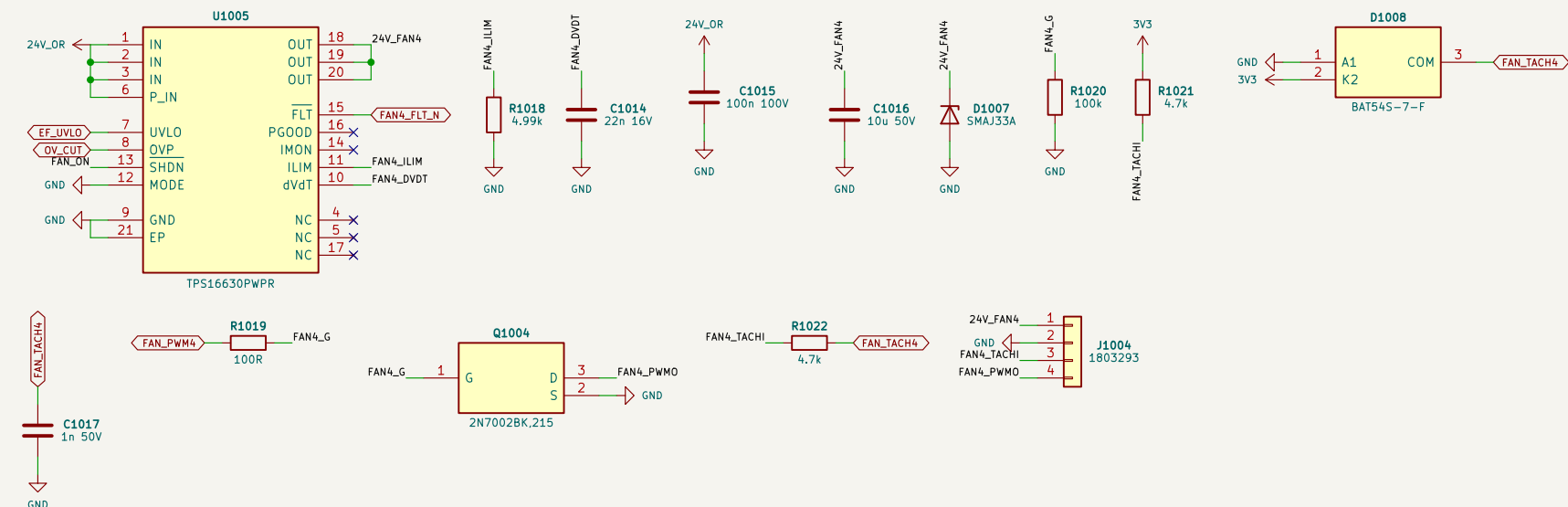
FAN 2



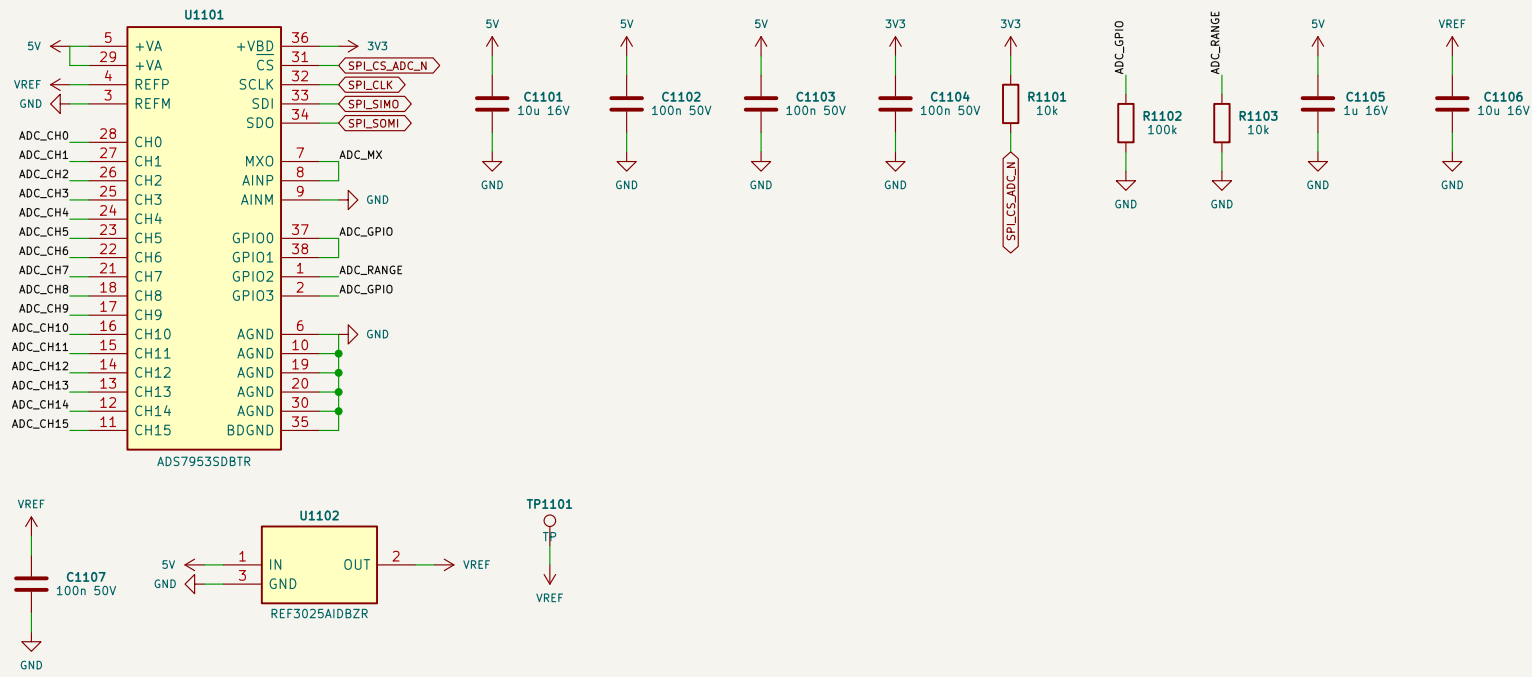
FAN 3



FAN 4

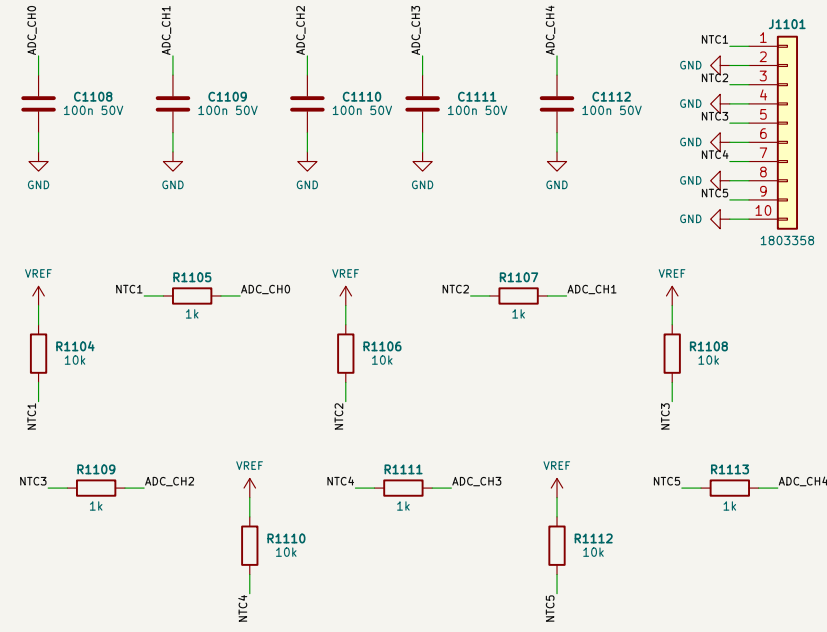


ADC AND REFERENCE



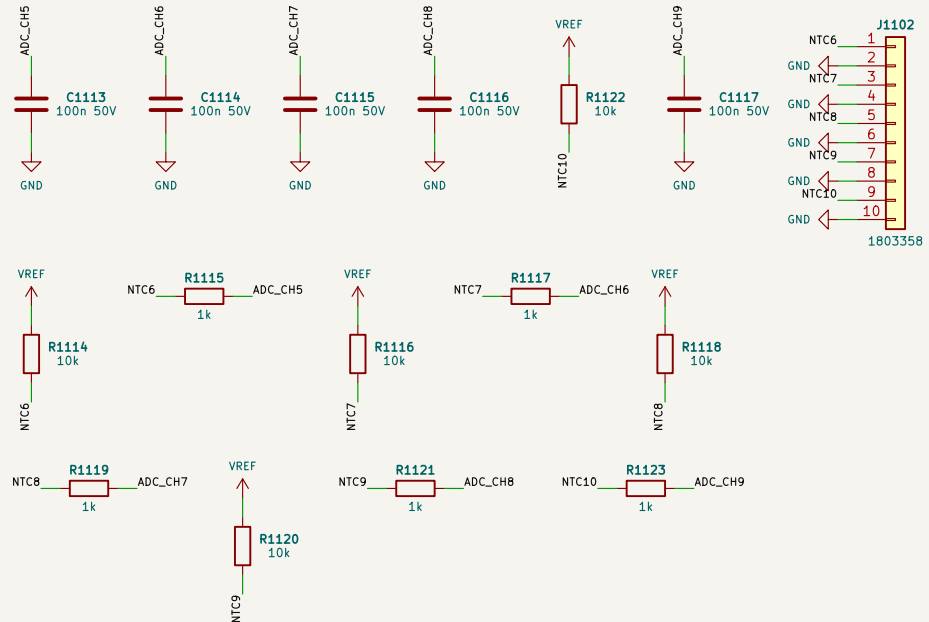
ADS7953: +VA 5 V, +VBD 3.3 V, range 1 (0..VREF, GPIO2 pulled low), MX0 -> AINP.
GPIO0/1/3 inputs by default, pulled low. SDO tri-states when CS is high.

NTC1-NTC5



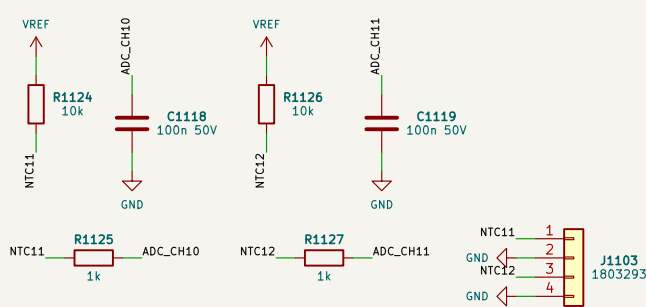
NTC: B57703M0103 (10k, B 3988), VREF 2.5 V -> 10k 0.1 % -> NTC: 1k + 100n -> ADC (ratiometric).
Codes (12 bit): -40 C 3978, 25 C 2048, 125 C 135, 150 C 74, Open > 4040, short < 40:
both are outside the -40..150 C band, so a broken or shorted sensor is distinguishable.

NTC6-NTC10



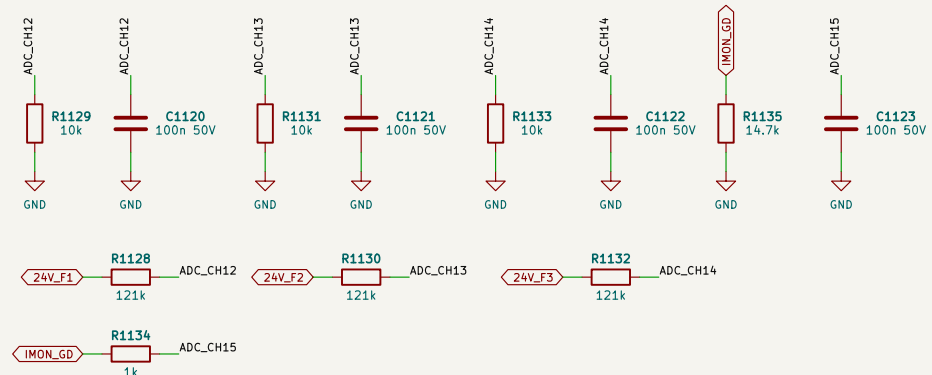
Terminal pairs: odd = NTC, even = 0 V return.

NTC11-NTC12



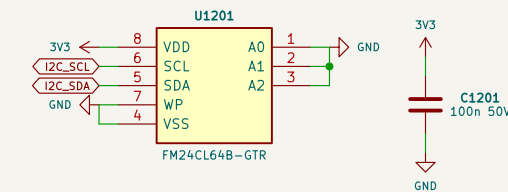
Terminal pairs: odd = NTC, even = 0 V return.

SUPPLY MONITORS



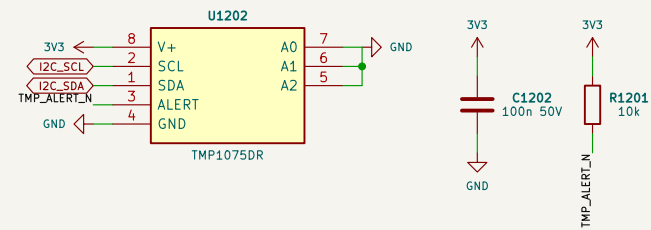
Feeds: 121k / 10k -> full scale 32.7 V (CH12 feed 1, CH13 feed 2, CH14 AUX-HV).
CH15: IMON_GD 27.9 uA/A x 14.7k -> 6 A = 2.46 V (gate-drive bias of all power boards).

SERVICE F-RAM



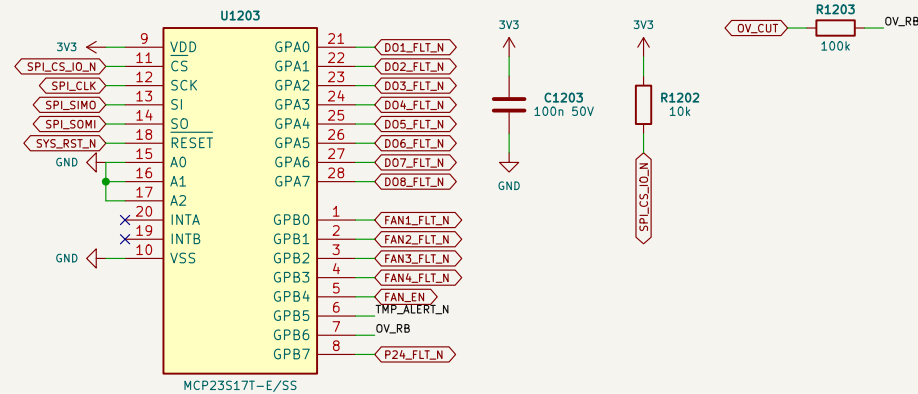
64 kbit F-RAM, 1e14 write cycles: an hours counter written every minute for 20 years is ~1e7 writes. WP low (writable). I2C pull-ups and the 0x57 EEPROM are on CTRL.

BOARD TEMPERATURE



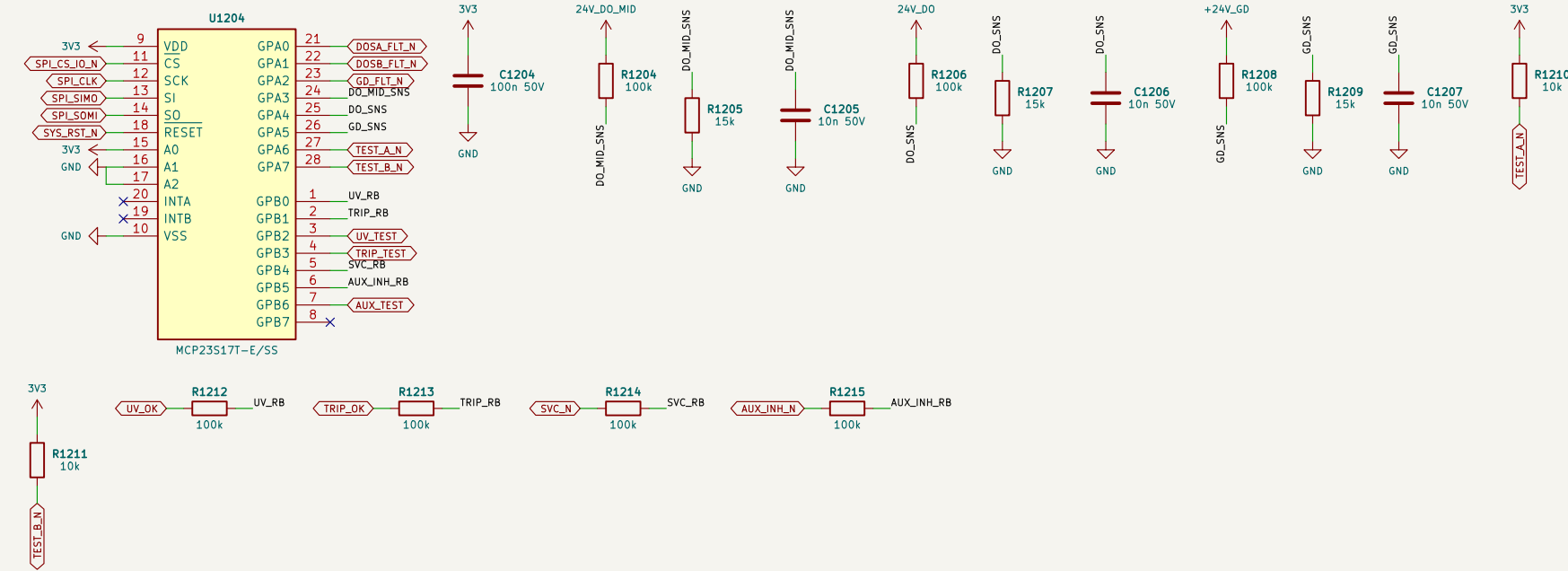
TMP1075 at 0x48 (no clash with 0x50 / 0x57); ALERT -> expander 1 GPB5

EXPANDER 1 (ADDRESS 0): OUTPUT-DRIVER AND FAN FAULTS



GP0-7 = D01-8 FLT (TPS272C45D), GPB0-3 = fan eFuse FLT, GPB4 = FAN_EN out, GPB5 = TMP ALERT, GPB6 = OV_CUT readback, GPB7 = +24V eFuse FLT, Inputs: enable GPPU pull-ups, Reset = SYS_RST_N, MCP23517 at 3.3 V is rated -40..+85 C (DS20001952C p1: +125 C only at 4.5-5.5 V): enough for PV-20

EXPANDER 2 (ADDRESS 1): CHAIN SELF-TEST AND SWITCH READBACK



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GPA0-2 = D05 A / D05 B / GD eFuse FLT, GPA3-5 = 24V_O_MID / 24V_DO_0 / +24V_GD present (100k/15k;
21.6 V ~> 2.82 V; source 3.93 V, clamp 0.1 mA), GPA6/7 = TEST_A_N / TEST_B_N outputs (10k pull-ups; idle healthy;
driving low can only trip a channel), GPB0-1 = UV_O_0 / TRIP_O_0 readback via 100k, GPB2/3 = UV_TEST /
TRIP_TEST outputs (100k pull-downs; idle off; high forces that detector input into its trip state),
GPB4 = SVC_RBL (service_jumper fitted = 0), GPB5 = AUX_INH_RBL (0 = AUX - HV is the source, power outputs
inhibited), GPB6 = AUX_TEST_jumper output (100k pull-up; high forces the inhibit), GPB7 spare (enable GPOU).

```